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BSIT3A

Advanced Data Communication

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Wireshark version 4.6.4 (Midterm Activity)

Answer #1

1. Detecting Slow Connections

Key Indicators:

- High latency (delay)
- TCP retransmissions
- Slow server responses

Steps using Wireshark 4.6.4:

A. Use Display Filter:

tcp.analysis.retransmission

If many packets appear -> network is slow or congested

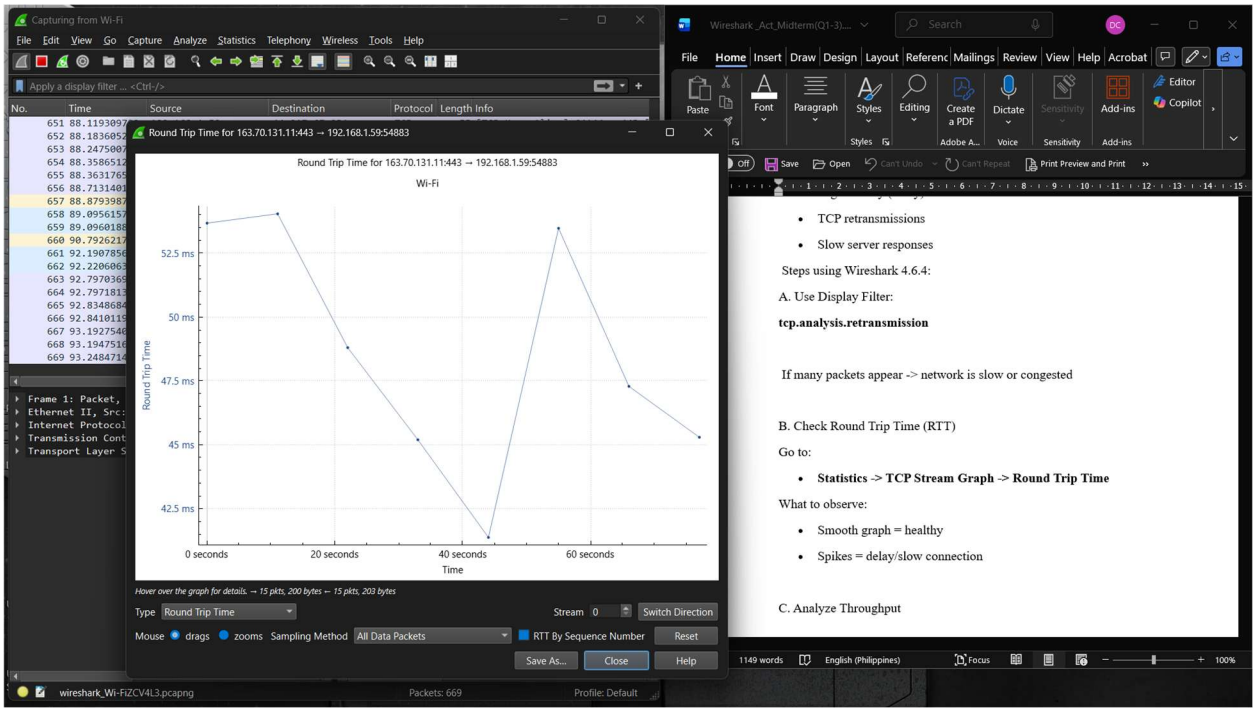
B. Check Round Trip Time (RTT)

Go to:

- **Statistics -> TCP Stream Graph -> Round Trip Time**

What to observe:

- Smooth graph = healthy
- Spikes = delay/slow connection



C. Analyze Throughput

Go to:

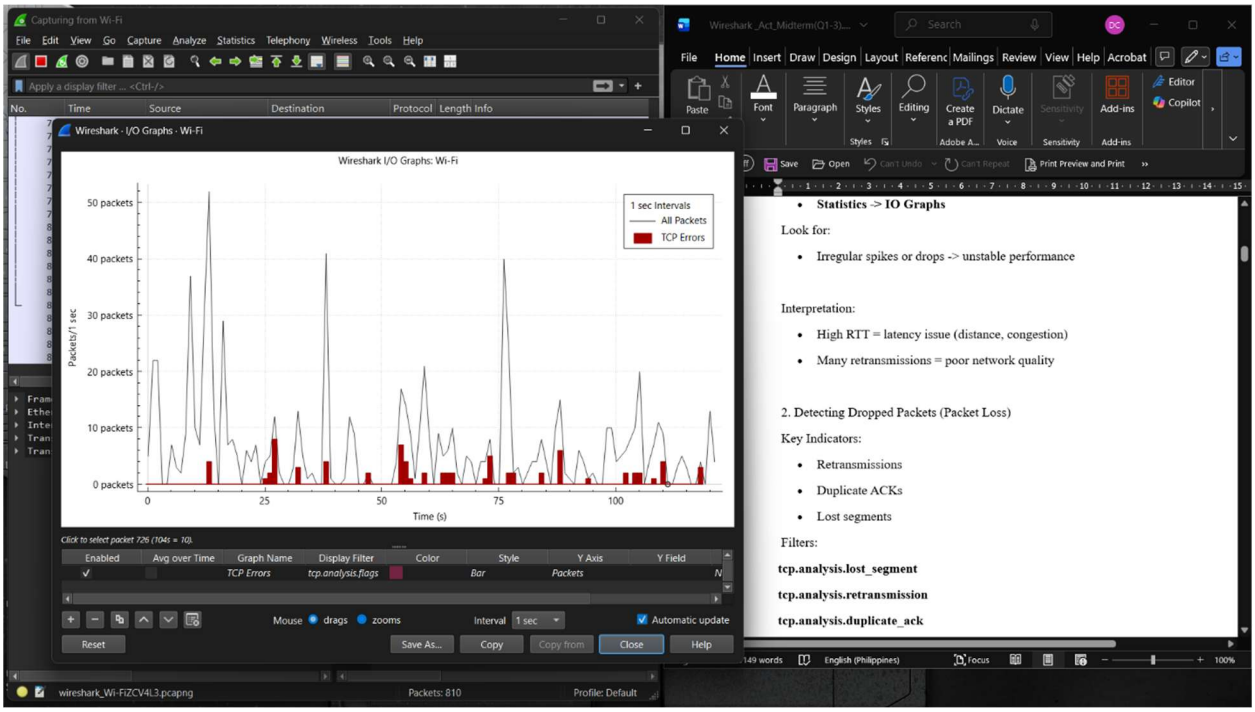
- **Statistics -> IO Graphs**

Look for:

- Irregular spikes or drops -> unstable performance

Interpretation:

- High RTT = latency issue (distance, congestion)
- Many retransmissions = poor network quality



2. Detecting Dropped Packets (Packet Loss)

Key Indicators:

- Retransmissions
- Duplicate ACKs
- Lost segments

Filters:

tcp.analysis.lost_segment

tcp.analysis.retransmission

tcp.analysis.duplicate_ack

Steps:

1. Capture traffic
2. Apply filters
3. Count occurrences

Interpretation:

- Lost segments -> packets never arrived
- Duplicate ACKs -> receiver is missing data
- Retransmissions -> sender is resending lost packets

Frequent occurrence = packet loss problem

3. Detecting Misconfigured Devices

Misconfigurations usually show abnormal or incorrect traffic patterns.

A. IP Address Conflicts (ARP Issues)

Filter:

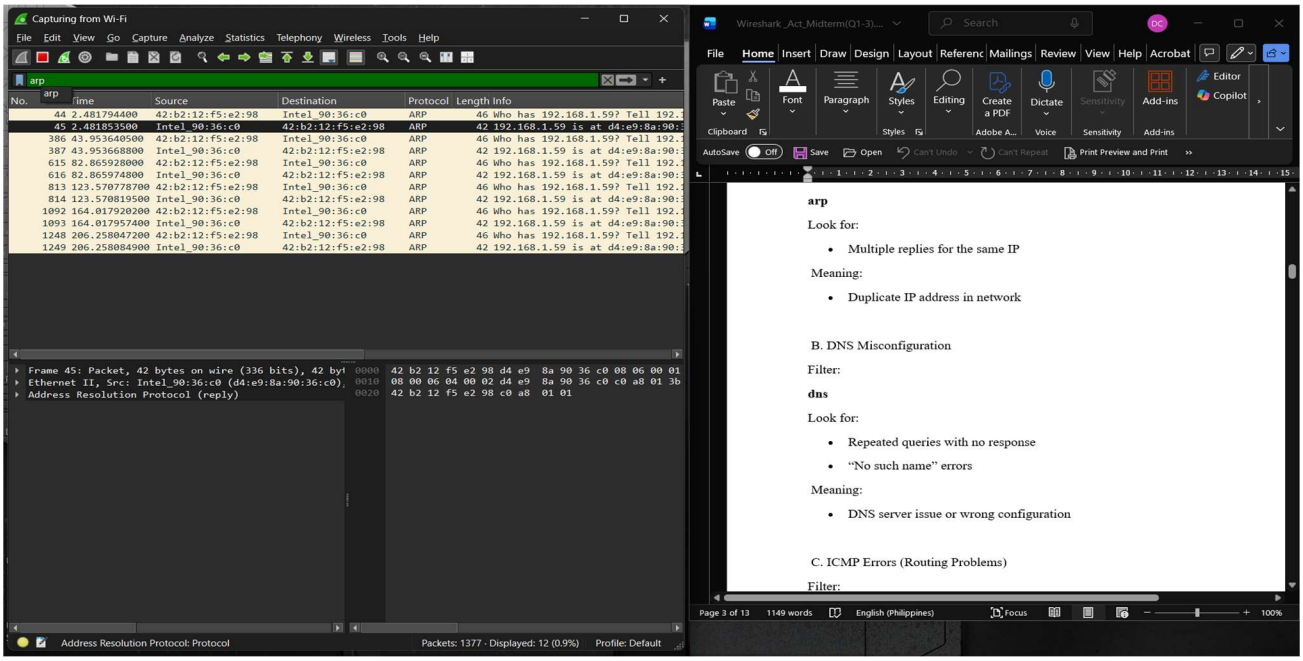
arp

Look for:

- Multiple replies for the same IP

Meaning:

- Duplicate IP address in network



B. DNS Misconfiguration

Filter:

dns

Look for:

- Repeated queries with no response
- “No such name” errors

Meaning:

- DNS server issue or wrong configuration

C. ICMP Errors (Routing Problems)

Filter:

icmp

Look for:

- Destination unreachable
- Time exceeded

Meaning:

- Incorrect routing or unreachable host

D. Incorrect Port/Service Usage

Filter examples:

tcp.port == 80

tcp.port == 443

Look for:

- Unexpected services running on the wrong ports

4. Use “Expert Information” (Best Feature in 4.6.4)

Go to:

- **Analyze -> Expert Information**

Wireshark will automatically flag:

- Errors (red)
- Warnings (yellow)
- Notes (blue)

Common findings:

- Retransmissions
- Duplicate ACKs
- Malformed packets

This is the fastest way to detect problems during exams or troubleshooting.

5. Additional Diagnostic Tools

Protocol Hierarchy

- **Statistics -> Protocol Hierarchy**
- Shows traffic distribution

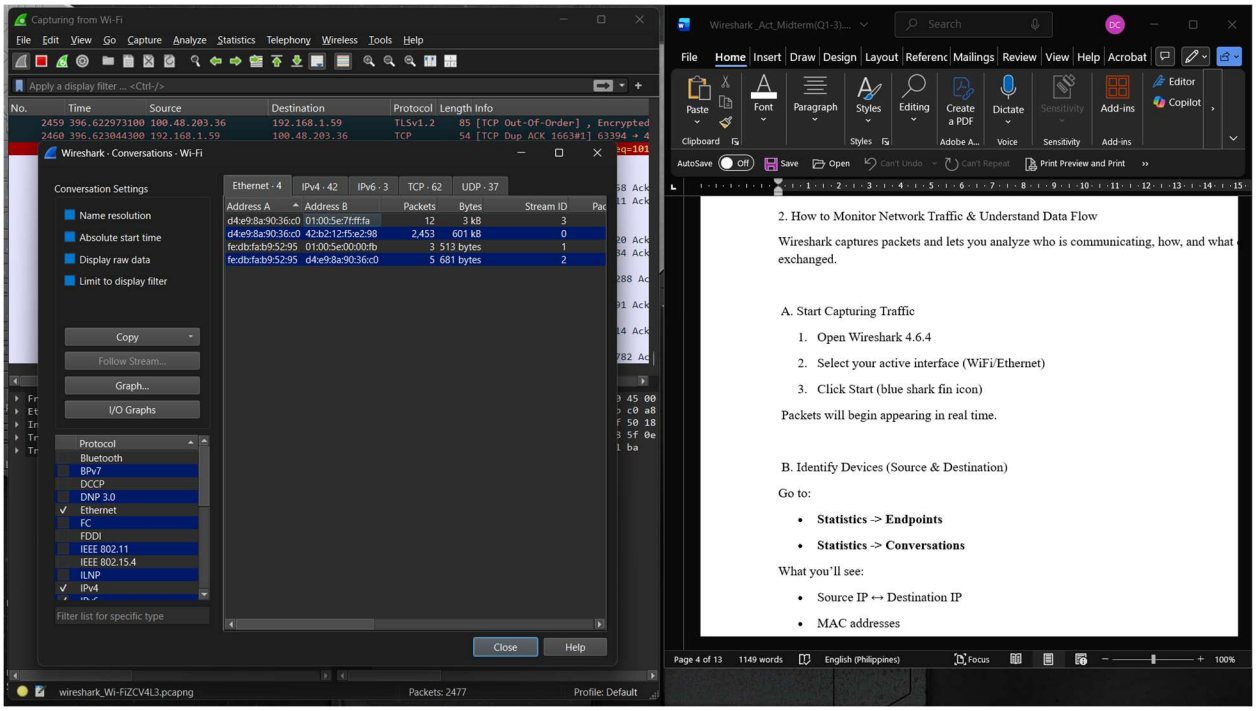
Detect unusual or unexpected protocols

Conversations

- **Statistics -> Conversations**

Helps identify:

- Which device is causing heavy traffic
- Communication patterns



Practical Example (Real Scenario)

If your network is slow:

- Apply tcp.analysis.retransmission
- Check IO Graphs for spikes
- Use Expert Info for warnings

If devices can't connect:

- **Check arp -> duplicate IP**
- **Check dns -> failed lookups**
- **Check icmp -> unreachable errors**

Summary Table

Problem	Indicator	Tool/Filter
Slow connection	High RTT, retransmissions	TCP Graphs, IO Graph
Packet loss	Lost segments, duplicate ACKs	tcp.analysis.*
Misconfiguration	ARP conflicts, DNS errors	arp, dns, icmp
General issues	Errors/warnings	Expert Information

Answer No. 2

Using Wireshark 4.6.4, we can both monitor how data flows across a network and detect suspicious or malicious activity.

2. How to Monitor Network Traffic & Understand Data Flow

Wireshark captures packets and lets you analyze who is communicating, how, and what data is exchanged.

A. Start Capturing Traffic

1. Open Wireshark 4.6.4
2. Select your active interface (WiFi/Ethernet)
3. Click Start (blue shark fin icon)

Packets will begin appearing in real time.

B. Identify Devices (Source & Destination)

Go to:

- **Statistics -> Endpoints**
- **Statistics -> Conversations**

What you'll see:

- Source IP ↔ Destination IP

- MAC addresses
- Bytes transferred

This helps you map:

“Device A communicates with Device B”

C. Understand Data Movement (Packet Structure)

Click any packet and observe 3 panes:

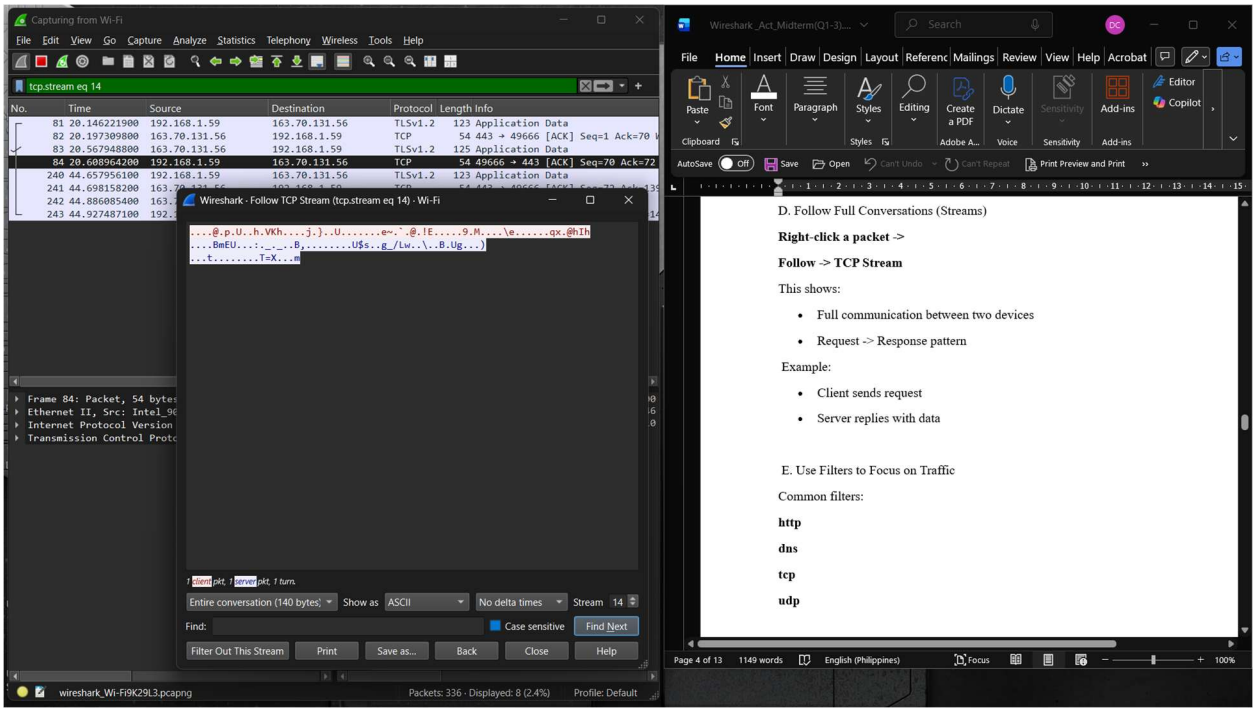
1. **Packet List Pane** – summary (time, source, destination, protocol)
2. **Packet Details Pane** – protocol layers:
    - Frame
    - Ethernet
    - IP
    - TCP/UDP
    - Application (HTTP, DNS, etc.)
3. **Packet Bytes Pane** – raw data

This shows how data moves through OSI layers

D. Follow Full Conversations (Streams)

Right-click a packet ->

Follow -> TCP Stream



This shows:

- Full communication between two devices
- Request -> Response pattern

Example:

- Client sends request
- Server replies with data

E. Use Filters to Focus on Traffic

Common filters:

http

dns

tcp

udp

icmp

Purpose:

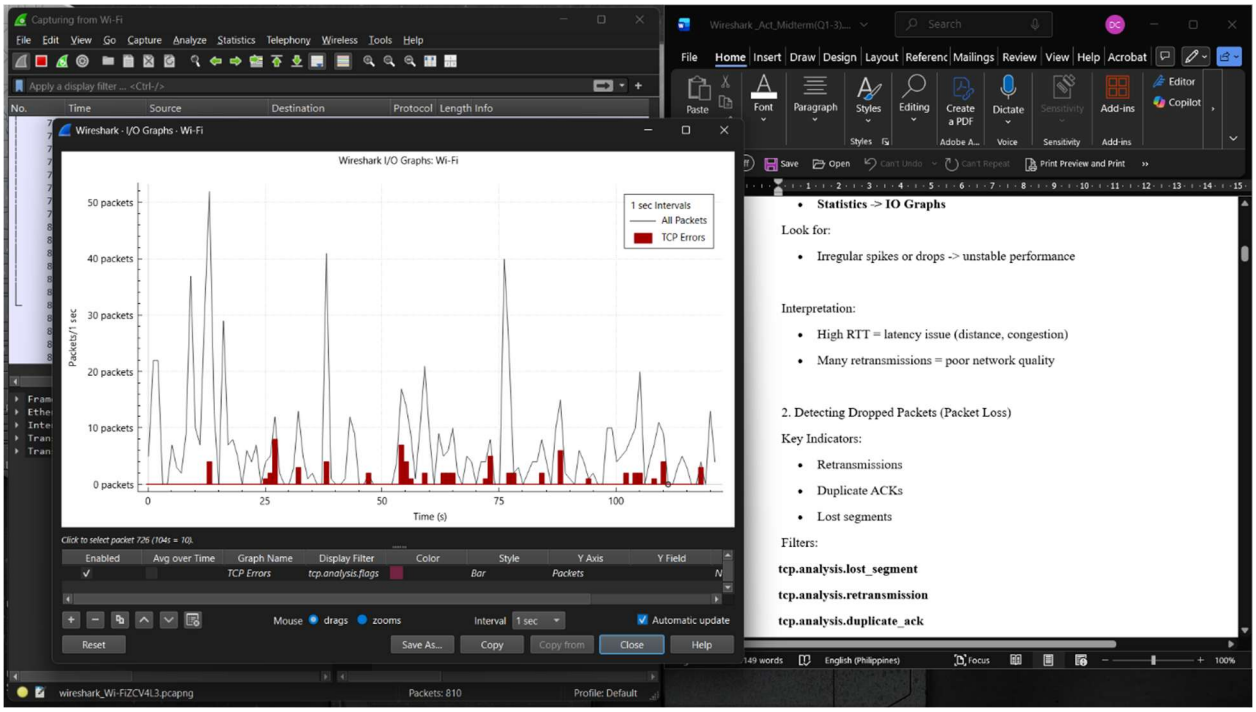
- **dns -> domain name lookups**

- **http -> web traffic (readable)**
- **tcp -> reliable communication**
- **udp -> fast communication**

F. Visualize Traffic Patterns

Go to:

- **Statistics -> IO Graphs**



Look for:

- Traffic spikes -> heavy usage
- Consistent flow -> normal activity

Key Concept (for explanation):

Data flow in a network:

Source -> Protocol -> Destination -> Response

Wireshark helps you analyze:

- Who initiated communication
- What protocol is used
- How fast data is exchanged

Answer No. 3

3. How to Identify Suspicious Activity, Attacks, or Malware

Wireshark is widely used in Cybersecurity to detect threats.

A. Unusual Traffic Patterns

Tool:

- **Statistics -> IO Graphs**

Signs:

- Sudden spikes in traffic
- Continuous high traffic

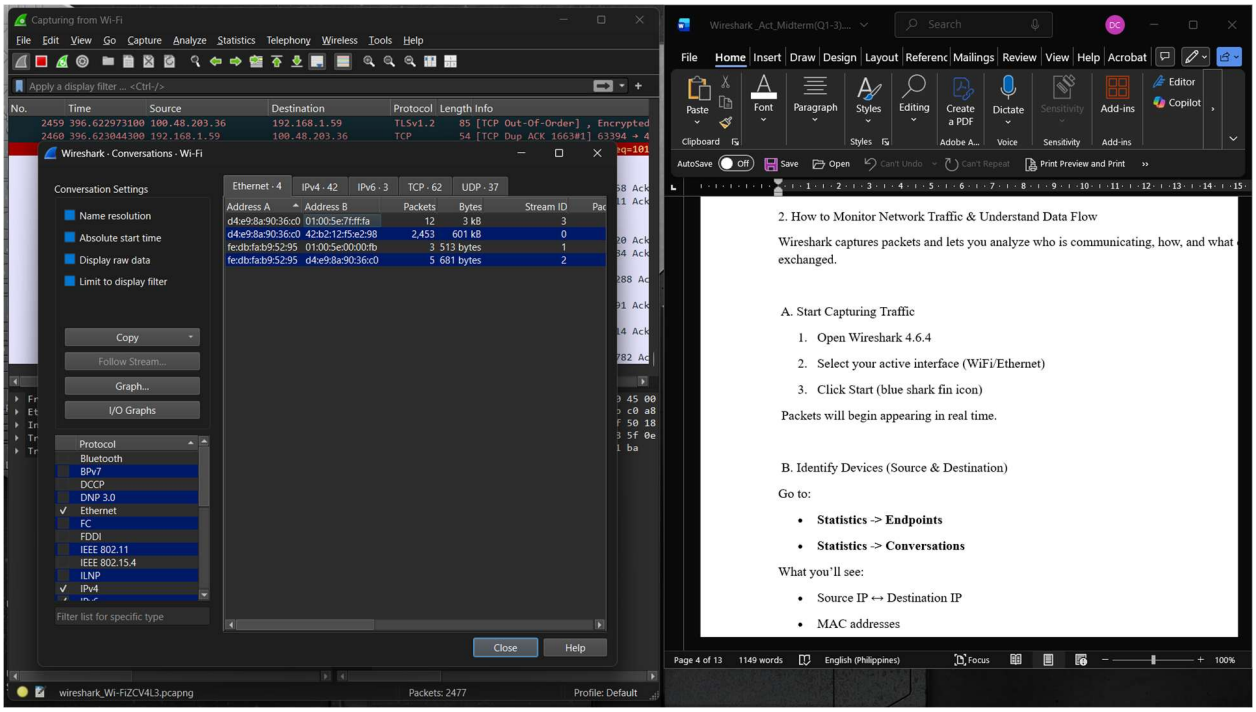
Possible:

- DDoS attack
- Malware activity

B. Unknown or Suspicious IP Communication

Go to:

- **Statistics -> Conversations**



Red flags:

- Internal device communicating with unknown external IP
- Repeated connections to same suspicious IP

C. Port Scanning (Reconnaissance Attack)

Filter:

tcp. flags.syn == 1 and tcp.flags.ack == 0

Meaning:

- Many SYN packets without completion

Indicates:

- Hacker scanning open ports

D. DNS-Based Suspicious Activity

Filter:

dns

Look for:

- Long/random domain names
- Repeated DNS queries

Possible:

- DNS tunneling (data exfiltration)

E. Malware Beaconing Behavior

Signs:

- Regular interval communication (e.g., every 10 seconds)
- Same destination IP

Meaning:

- Infected device contacting command server

F. Unencrypted Sensitive Data

Filter:

http

Look for:

- Plaintext credentials

Risk:

- Data exposure

G. ICMP Flood (Ping Attack)

Filter:

icmp

Look for:

- Large volume of ICMP packets

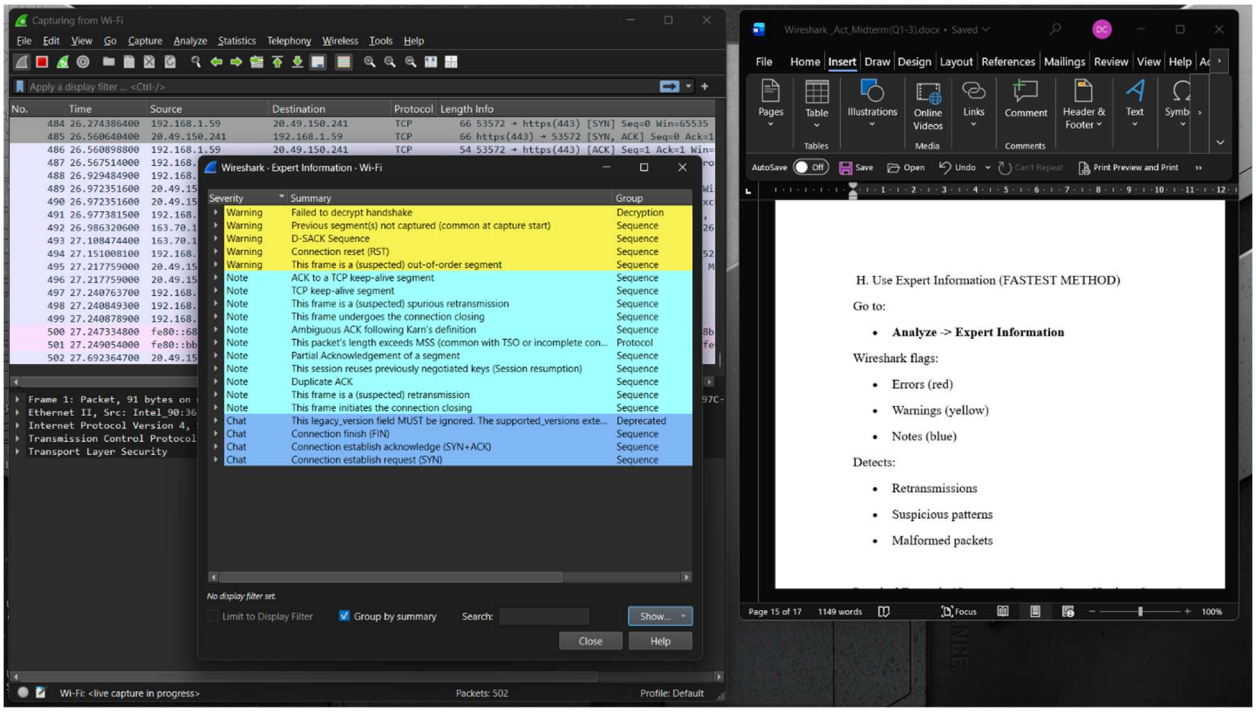
Possible:

- Denial-of-Service (DoS)

H. Use Expert Information (FASTEST METHOD)

Go to:

- **Analyze -> Expert Information**



Wireshark flags:

- Errors (red)
- Warnings (yellow)
- Notes (blue)

Detects:

- Retransmissions
- Suspicious patterns
- Malformed packets

Practical Example (Capstone Context_ Smart Hygiene System)

For your IoT system:

- If ESP32 sends continuous traffic -> possible bug or attack

- If it connects to an unknown IP -> possible compromise
- If DNS queries are abnormal -> possible malware

Summary Table

Task	Tool	What to Look For
Monitor traffic	Conversations	Device communication
Understand flow	TCP Stream	Request/response
Analyze protocols	Filters	http, dns, tcp
Detect attacks	IO Graphs	Traffic spikes
Identify malware	Patterns	Repeated connections
Quick diagnosis	Expert Info	Errors/warnings

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